

1. Learning contracts and loss

1. Weak versus strong learning

In a declared realizable binary-classification scope, a **weak PAC learner** has some fixed edge $\gamma > 0$ and, for every allowed population distribution and confidence request δ , returns with probability at least $1 - \delta$ a hypothesis with population error at most

$$\frac{1}{2} - \gamma,$$

using the declared polynomial resources. A **strong PAC learner** meets every requested accuracy $\varepsilon > 0$ and confidence $1 - \delta$ in the declared PAC setting, with the relevant polynomial resource dependence.

Inside AdaBoost, the separate roundwise empirical condition is that the base hypothesis has weighted error below $\frac{1}{2}$ under the current normalized weights of the fixed training sample. One observed accuracy does not establish the population weak-PAC guarantee. If the current error is $e < \frac{1}{2}$, its observed advantage is $\frac{1}{2} - e$; a uniform rate needs a positive lower bound that holds in every relevant round.

2. Example distribution and exponential loss

At round t , D_t is a positive normalized distribution on the fixed training indices:

$$D_{t(i)} > 0, \quad \sum_{i=1}^m D_{t(i)} = 1.$$

It is not the unknown population distribution in a PAC guarantee. For binary $y \in \{-1, +1\}$ and a real score $F(x)$, the **exponential loss** is

$$\ell_{\exp}(y, F(x)) = e^{-yF(x)}.$$

Writing $z = yF(x)$ and $\varphi(z) = e^{-z}$, we have $\varphi''(z) = e^{-z} > 0$, so the loss is infinitely differentiable and convex. It is **not globally L -smooth on $\mathbb{R}$** : $\varphi''(z)$ is unbounded as $z \rightarrow -\infty$. On $[-B, B]$ it is e^B -smooth. Independently, it pointwise upper-bounds the classification-error indicator:

$$1_{[yF(x) \leq 0]} \leq e^{-yF(x)}.$$

This is a surrogate-loss bound, not an observed error rate.

Differentiability, convexity, global smoothness, the indicator bound, a training rate, and test performance are distinct properties.

2. One exact AdaBoost round

1. Weighted error and coefficient

For a binary candidate h_t , first measure its weighted error under the already normalized D_t :

$$\varepsilon_t = \Pr_{i \sim D_t} [h_{t(x_i)} \neq y_i] = \sum_{i: h_{t(x_i)} \neq y_i} D_{t(i)}, \quad 0 < \varepsilon_t < \frac{1}{2}.$$

For a scalar coefficient a , the roundwise exponential objective is

$$\Phi_{t(a)} = (1 - \varepsilon_t)e^{-a} + \varepsilon_t e^a.$$

Its derivative vanishes exactly when $e^{2a} = \frac{1 - \varepsilon_t}{\varepsilon_t}$. Since $\Phi_{t''}(a) > 0$, the unique finite minimizer is

$$\alpha_t = \frac{1}{2} \log \frac{1 - \varepsilon_t}{\varepsilon_t} > 0.$$

As ε_t approaches $\frac{1}{2}$ from below, α_t approaches 0: a near-random hypothesis receives almost no ensemble weight.

2. Normalize the signed update

Form the unnormalized weights and their total:

$$\begin{aligned} \tilde{D}_{t+1}(i) &= D_{t(i)} e^{-\alpha_t y_i h_{t(x_i)}}, \\ Z_t &= \sum_{j=1}^m D_{t(j)} e^{-\alpha_t y_j h_{t(x_j)}} > 0. \end{aligned}$$

Then normalize:

$$D_{t+1}(i) = \frac{D_{t(i)} e^{-\alpha_t y_i h_{t(x_i)}}}{Z_t}.$$

A correctly classified row has factor $e^{-\alpha_t} < 1$; an incorrect row has factor $e^{\alpha_t} > 1$. These are relative changes and become a probability distribution only after division by Z_t . Under exact binary minimization, the total correct and incorrect mass each become $\frac{1}{2}$ in D_{t+1} , while rows within each group retain their previous relative weights.

3. Complete algorithm and output type

Initialize $D_1(i) = \frac{1}{m}$. For $t = 1, \dots, T$:

1. obtain h_t using D_t ;
2. compute ε_t and require $0 < \varepsilon_t < \frac{1}{2}$ for this derivation;
3. set $\alpha_t = \frac{1}{2} \log \left(\frac{1 - \varepsilon_t}{\varepsilon_t} \right)$;
4. compute Z_t and the normalized update above.

The real ensemble score and binary prediction are

$$F_{T(x)} = \sum_{t=1}^T \alpha_t h_{t(x)}, \quad \text{sign}(F_{T(x)}).$$

The score accumulates evidence; its sign is the label. Do not return F_T itself as a binary label, omit Z_t , or replace the log-odds coefficient by an ad hoc function of ε_t .

4. One exact update to recognize

Let $D_t = (0.10, 0.25, 0.25, 0.40)$ and suppose only row 2 is misclassified. Then

$$\varepsilon_t = \frac{1}{4}, \quad \alpha_t = \frac{1}{2} \log 3,$$

so correct rows receive $e^{-\alpha_t} = \frac{1}{\sqrt{3}}$ and row 2 receives $e^{\alpha_t} = \sqrt{3}$. The unnormalized weights are

$$\left(\frac{0.10}{\sqrt{3}}, 0.25\sqrt{3}, \frac{0.25}{\sqrt{3}}, \frac{0.40}{\sqrt{3}} \right),$$

with $Z_t = \frac{\sqrt{3}}{2}$. After normalization,

$$D_{t+1} = \left(\frac{1}{15}, \frac{1}{2}, \frac{1}{6}, \frac{4}{15} \right).$$

They sum to one; the previously incorrect row has half the mass and the three correct rows together have the other half.

3. Training-error certificate

1. Exact telescoping identity

Unrolling the normalized update for one index gives

$$D_{T+1}(i) = \frac{\exp(-y_i F_{T(x_i)})}{m \prod_{t=1}^T Z_t}.$$

Using $\sum_i D_{T+1}(i) = 1$ yields the exact identity

$$\frac{1}{m} \sum_{i=1}^m \exp(-y_i F_{T(x_i)}) = \prod_{t=1}^T Z_t.$$

The ordinary unweighted average appears because $D_1(i) = \frac{1}{m}$. For a normalized but nonuniform D_1 , the left side becomes $\sum_i D_1(i) e^{-y_i F_{T(x_i)}}$. This is a training-sample identity, not a population expectation.

2. Exact factor and edge bounds

Splitting correct and incorrect D_t -mass and inserting the registered α_t gives

$$Z_t = (1 - \varepsilon_t) e^{-\alpha_t} + \varepsilon_t e^{\alpha_t} = 2\sqrt{\varepsilon_t(1-\varepsilon_t)}.$$

With the round edge $\gamma_t = \frac{1}{2} - \varepsilon_t$,

$$Z_t = \sqrt{1 - 4\gamma_t^2} \leq e^{-2\gamma_t^2},$$

where the inequality follows from $1 - x \leq e^{-x}$; it is not a small-edge approximation. Therefore

$$\hat{R}_{S(\text{sign}(F_T))} \leq \prod_{t=1}^T Z_t = \prod_{t=1}^T 2\sqrt{\varepsilon_t(1-\varepsilon_t)} \leq \exp\left(-2 \sum_{t=1}^T \gamma_t^2\right).$$

If one fixed $\gamma > 0$ satisfies $\gamma_t \geq \gamma$ in every round, then

$$\hat{R}_{S(\text{sign}(F_T))} \leq \exp(-2\gamma^2 T).$$

All right-hand quantities are upper bounds on observed empirical error —not the observed error itself and not test-error guarantees.

3. What makes the proof work

The logical chain is

$$1_{[y_i F_{T(x_i)} \leq 0]} \leq e^{-y_i F_{T(x_i)}} \rightarrow \prod_t Z_t \rightarrow \exp\left(-2 \sum_t \gamma_t^2\right).$$

It uses, in order: the pointwise indicator bound; the exact telescoping identity from normalized updates; the exact factor $Z_t = 2\sqrt{\varepsilon_t(1-\varepsilon_t)}$; the round edges; and $1 - x \leq e^{-x}$. Convexity alone supplies neither this empirical rate nor a population conclusion. The fixed-edge corollary

may be used only after one common positive lower bound has been established for every relevant round.

4. After zero training error

1. Signs do not determine score magnitudes

Zero empirical classification error establishes only

$$y_i F_{T(x_i)} > 0 \quad \text{for every training example.}$$

It does not determine the magnitudes or distribution of these signed scores. Later valid rounds can still change F_T and move them. Test or validation error may decrease, flatten, or increase; zero training error alone rules out none of these behaviors.

Continued test improvement after zero training error is a possible empirical pattern, not a universal theorem, a monotonicity guarantee, a fixed optimal round count, or a rule to continue forever. Noise, persistent hard examples, an insufficient base-hypothesis class, and repeated use of validation data can change the outcome. Test behavior needs separate evidence.

2. Three complementary views

View	Inspect	Supports
algorithmic	$D_t \rightarrow D_{t+1}$	which examples influence the next learner call; normalization invariant
roundwise optimization	$\varepsilon_t \rightarrow \alpha_t$	why the exact coefficient minimizes the one-dimensional empirical exponential objective
signed-score evolution	$y_i F_{t(x_i)}$	what the zero-one training count discards

The selected h_t and weighted error connect the first two views; $\alpha_t h_t$ connects optimization to the update of F_t . Shared state does not make the conclusions interchangeable. Reweighting is an implementation statement, exact line minimization is an empirical objective statement,

and score evolution is a diagnosis. None alone is a population generalization theorem.

3. Fast claim audit

- **Population vs sample:** D_t is on fixed training indices; $\hat{R}_S$ and the product bound are empirical.
- **Bound vs observation:** $\prod_t Z_t$ and the edge expressions certify an upper bound; they are not measured error values.
- **Score vs prediction:** F_T is real-valued; $\text{sign}(F_T)$ is the label.
- **Local vs global edge:** $\gamma_t = \frac{1}{2} - \varepsilon_t$ is round-specific; a fixed γ requires an all-round lower-bound condition.
- **Training vs generalization:** normalized updates, loss minimization, score movement, and even zero training error do not by themselves imply low future test error.

4. Compare three valid certificates

For $\varepsilon_1 = 0.30$, $\varepsilon_2 = 0.25$, and $\varepsilon_3 = 0.40$, the edges are 0.20, 0.25, 0.10.

The exact-factor certificate is

$$\sqrt{0.84}\sqrt{0.75}\sqrt{0.96} \approx 0.7778.$$

Using every round edge gives

$$\exp(-2(0.20^2 + 0.25^2 + 0.10^2)) = \exp(-0.225) \approx 0.7985.$$

Using only the minimum common edge $\gamma = 0.10$ gives

$$\exp(-2(0.10)^2(3)) = \exp(-0.06) \approx 0.9418.$$

Here the exact product is tightest, the varying-edge bound is next, and the fixed-edge corollary is loosest. All three remain empirical-error certificates, not observations.